

SatQuery AI | Earth Observation Evaluation

MISSION QUERY & TASK SPECIFICATION

Natural Language Query:	"What changed between these two dates?"
Classified Task Type:	Change Detection (Bi-temporal)
Confidence Metric:	92% (Dual-Level Verification: Model + Spatial Evidence)
Execution Status:	COMPLETED (Deterministic DAG Verified)

EXECUTIVE AI ANALYSIS & REASONING SUMMARY

Primary Finding: Built-up area increased in the eastern section, mainly around the new road corridor and adjacent settlements.

- Significant increase in built-up structures (red) near the main road.
- New construction observed along the eastern cluster.
- No significant change in vegetation along the river.
- Some reduction in bare land in the southern region.

GEOSPATIAL & SENSOR SPECIFICATIONS

Parameter	Specification	Parameter	Specification
Sensor 1 (T1)	Sentinel-2 MSI L2A	Observation Date (T1)	2022-01-15
Sensor 2 (T2)	Sentinel-2 MSI L2A	Observation Date (T2)	2024-06-20
CRS / Projection	EPSG:4326 (WGS 84)	Spatial Resolution	10.0 meters / pixel
Scene Footprint	10 km x 10 km (100 km²)	Center Coordinates	19.0760° N, 72.8777° E

AUDITABLE AGENTIC EXECUTION WORKFLOW

Step	Phase	Engine / Model	Validation / Rationale
01	Query Understanding	Agentic Query Router	Semantic classification to bi-temporal change task
02	Input Validation	Raster & Co-Registration Validator	Verified CRS match, 100% overlap, 10m GSD
03	Model Selection	Dynamic Specialist Registry	Bound RS-ChangeNet & CDVQA specialist tools
04	Change Extraction	RS-ChangeNet (Siamese)	Morphological segmentation, change clusters isolated
05	Evidence Integration	Evidence Fusion Engine	Calculated 92% confidence, 14.2% change area
06	Response Generation	Mission Report Generator	Synthesized visual overlays & PDF/JSON telemetry